What is AWS Lambda?

AWS Lambda is a cloud service provided by Amazon Web Services (AWS) that runs your code **automatically** without needing you to manage any servers.

Think of it like this:

- Imagine you have some **code** that performs a task (e.g., sending an email).
- AWS Lambda takes your code, runs it **when needed**, and stops when the task is done.
- You don't need to worry about buying or setting up a computer/server.

Real-World Example:

Imagine you're running an **e-commerce app** like Amazon.

Task:

When a customer places an order, you want to send a **thank you email**.

Here's how AWS Lambda works:

1. The customer places an order → **This is an event**.
2. AWS Lambda **automatically runs** your code to send the email.
3. Once the email is sent, AWS Lambda **stops running**.

You didn't have to:

- Buy a server.
- Keep the server running all the time.
- Pay for the server when it's not being used.

Key Points to Understand AWS Lambda:

1. **Serverless**:
   You don't need to manage any servers. AWS handles all that for you.

2. **Pay Only When It Runs**:
   You are charged only when your code runs. If your code runs for 2 seconds, you pay only for that.

3. **Event-Driven**:
   AWS Lambda runs your code in response to certain **events**.
   For example:

   - A **file is uploaded** to Amazon S3 (a storage service).
   - A user clicks a button on your app.
   - A new record is added to a database.
4. **Code Only**:
   You write the code, upload it to Lambda, and AWS takes care of the rest (like running it).

How to Use AWS Lambda in Simple Steps:

1. Write your code (e.g., Python, JavaScript).
2. Upload the code to AWS Lambda.
3. Tell AWS Lambda **when to run your code** (e.g., when a file is uploaded).
4. AWS runs the code when the event happens.

Benefits of AWS Lambda:

1. **No Server Management** → AWS handles everything.
2. **Cost-Effective** → You pay only for the time the code runs.
3. **Scalable** → AWS can handle 1 task or 1 million tasks.
4. **Easy Integration** → Works with other AWS services like S3, DynamoDB, etc.

What is Job Scheduling?

Job scheduling is like setting an **alarm clock**:

- You tell it **when** to ring.
- The alarm rings at the right time, and the task happens.

In AWS Lambda, you can schedule jobs to run your code **at specific times** using **EventBridge**.

How Does It Work?

1. **You create a schedule** (like setting an alarm).
2. AWS Lambda **automatically runs your code** at the time you set.
3. After the task is done, Lambda stops.

Example 1: Running Code Every Day

Let’s say you want to clean up some old files in your storage every day at **12:00 AM**.

1. Write a Lambda function to delete old files.
2. Create a schedule using **EventBridge** to trigger Lambda daily at 12:00 AM.
3. Lambda does the cleanup task every day, and you don’t need to worry about it!

How to Schedule Jobs:

1. **Go to AWS Lambda**:
   - Create a Lambda function with your code.
2. **Set a Schedule**:
   - Use **EventBridge** (formerly CloudWatch Events) to create a "rule."
   - Define when the function should run (e.g., every 5 minutes, daily, weekly).
3. **Lambda Executes**:
   - Your function will automatically run according to the schedule.

Cron Expressions for Scheduling

AWS uses **cron expressions** to define schedules.
Here's an example of a cron expression:

- `0 9 * * ? *` → Runs every day at 9:00 AM.

Breakdown:

- `0` → Minute (0th minute).
- `9` → Hour (9 AM).
- `*` → Any day of the month.
- `*` → Any month.
- `?` → No specific day of the week.
- `*` → Every year.

Key Benefits of Job Scheduling in Lambda:

1. **Automatic** → You don't need to remember to run the code.
2. **No Servers** → AWS manages everything.
3. **Cost-Effective** → You only pay when the job runs.